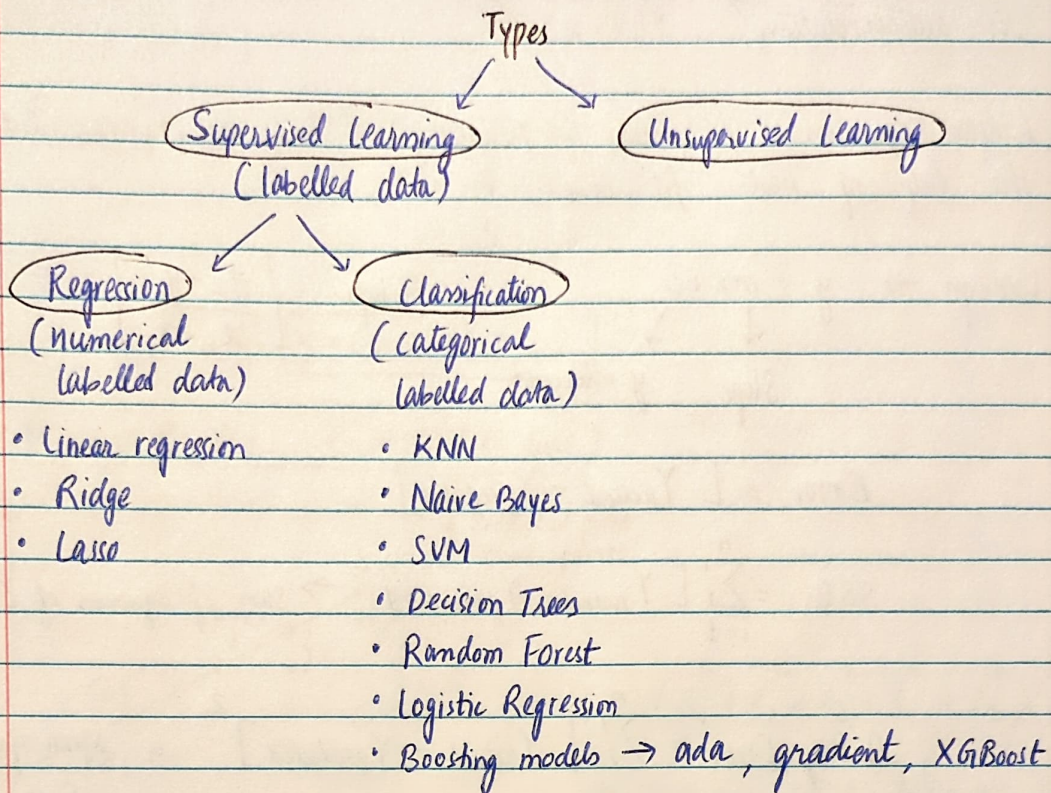


MACHINE LEARNING



1) Linear Regression (to predict numerical values)

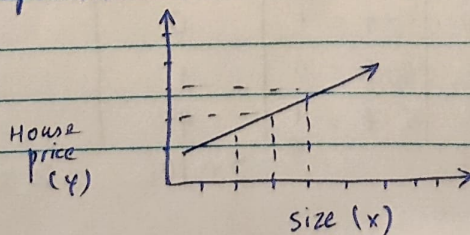
	Size (thousand sq.ft)	no of rooms	age of house	House price (Lakhs)
1	12	5	3	40
2	15	8	2	50
3	7	3	10	25
4	6	3	19	22
5	25	10	3	75
6	30	15	1	80
7	18	9	2	60
	x_1	x_2	x_3	y

independent columns
(x)

Target
(dependent)

Assumptions

(i) Each independent column should be linear to target column



- (iv) No Heteroskedasticity / it should be homoskedastic
- (ii) No multicollinearity between independent columns
- (iii) No autocorrelation

- Linear Regression is a process of predicting a numerical value with the help of best fit line.

Line eqn $\rightarrow y = ma + c$

$\downarrow$ $\downarrow$
 Slope y intercept

$$\text{Slope} = \left[\frac{y_2 - y_1}{x_2 - x_1} \right]$$

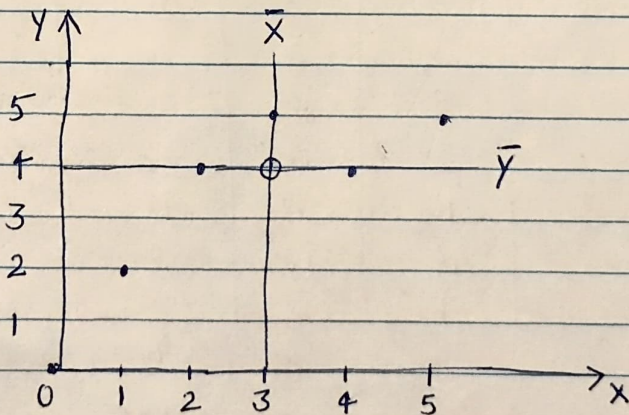
$$\text{Error} = [Y_{\text{actual}} - Y_{\text{predicted}}]$$

$$\text{SSE} = \sum_{i=1}^n [Y_{\text{Actual}} - Y_{\text{predicted}}]^2 \rightarrow \text{Sum of Squares of Errors}$$

$$\text{Average Error} = \frac{\sum_{i=1}^n [Y_{\text{actual}} - Y_{\text{predicted}}]^2}{n} \rightarrow \text{Mean squared Error}$$

$$\text{Root mean Squared error} = \sqrt{\frac{\sum_{i=1}^n [Y_{\text{actual}} - Y_{\text{predicted}}]^2}{n}}$$

Week	Cars sold
1	2
2	4
3	5
4	4
5	5
mean = 3 ($\bar{x}$)	mean = 4 ($\bar{y}$)



$$\text{mean} = \frac{\sum n}{n}$$

x	y	$x - \bar{x}$	$y - \bar{y}$	$(x - \bar{x})^2$	$(x - \bar{x})(y - \bar{y})$
1	2	$1 - 3 = -2$	$2 - 4 = -2$	$(-2)^2 = 4$	4
2	4	$2 - 3 = -1$	$4 - 4 = 0$	1	0
3	5	$3 - 3 = 0$	$5 - 4 = 1$	0	0
4	4	$4 - 3 = 1$	$4 - 4 = 0$	1	0
5	5	$5 - 3 = 2$	$5 - 4 = 1$	4	2

$$\Sigma = 10$$

$$\Sigma = 6$$

$$y = mx + c$$



$$m = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sum (x - \bar{x})^2} = \frac{6}{10} = 0.6$$

$$\bar{y} = m\bar{x} + c \Rightarrow c = \bar{y} - m\bar{x}$$

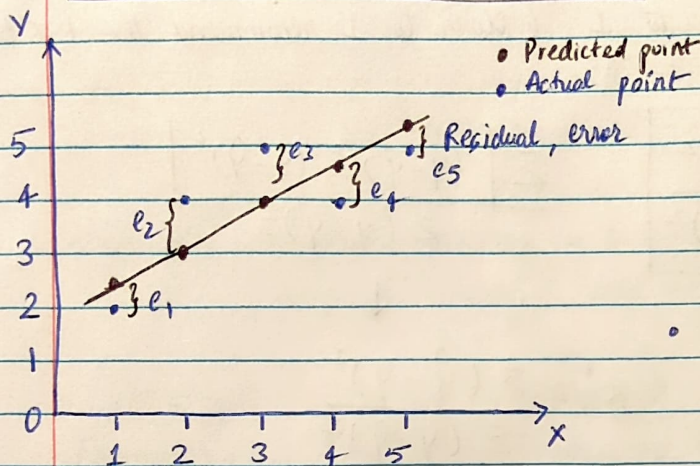
$$c = 4 - (0.6) \cdot 3$$

$$= 4 - 1.8$$

$$c = 2.2$$

$$\therefore \boxed{y = 0.6(x) + 2.2}$$

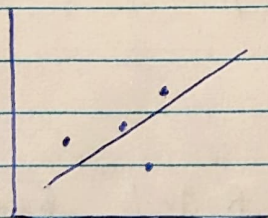
Minimal Residue (best fit line)



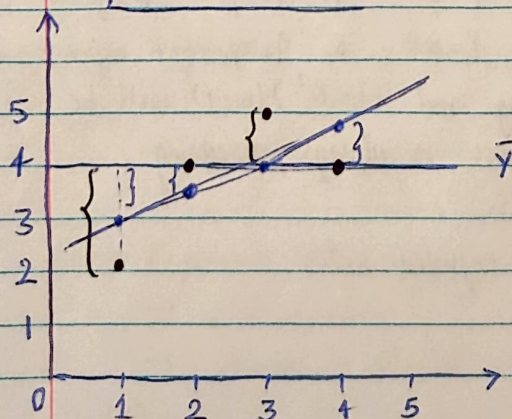
$$(e_1^2 + e_2^2 + e_3^2 + e_4^2 + e_5^2) = \text{smallest error}$$

- But here e_1, e_4 & e_5 are in the negative direction & e_2, e_3 are in positive direction
- if we add them they get cancelled
Therefore we take the squares of each error.
- We rotate the line to get the minimal error (best fit line)

R^2 - it tells us how well the line predicts actual points



- $\rightarrow$ this is an outlier
It is essential to remove the outliers,
this process is essential in EDA



Compare
distance b/w (actual - mean) distance b/w (estimated - mean)

- ① $\bar{y}$
- ② Actual data point
- ③ predicted line

		(Actual - mean)		(estimated - mean)		
X	Y	$y - \bar{y}$	$(y - \bar{y})^2$	$\hat{y} = m(x) + c$	$\hat{y} - \bar{y}$	$(\hat{y} - \bar{y})^2$
1	2	$2 - 4 = -2$	4	$0.6(1) + 2.2 = 2.8$	$2.8 - 4 = -1.2$	1.44
2	4	$4 - 4 = 0$	0	$0.6(2) + 2.2 = 3.4$	$3.4 - 4 = -0.6$	0.36
3	5	$5 - 4 = 1$	1	$0.6(3) + 2.2 = 4$	$4 - 4 = 0$	0
4	4	$4 - 4 = 0$	0	$0.6(4) + 2.2 = 4.6$	$4.6 - 4 = 0.6$	0.36
5	5	$5 - 4 = 1$	1	$0.6(5) + 2.2 = 5.2$	$5.2 - 4 = 1.2$	1.44
			$\Sigma = 6$			
						$\Sigma = 3.6$

$$R^2 = \frac{\sum (\hat{y} - \bar{y})^2}{\sum (y - \bar{y})^2} = \frac{3.6}{6} = 0.6$$

Range of $R^2 = 0$ to 1 (closer to 1 recommend the best fit line)

$$R^2 = 1 - \frac{\sum (y - \hat{y})^2}{\sum (y - \bar{y})^2} = \frac{\sum [(y - \bar{y})^2 - (y - \hat{y})^2]}{\sum (y - \bar{y})^2}$$

$$R^2 = 1 - \frac{SS_{\text{Residual}}}{SS_{\text{Total}}}$$

$$R^2 = \frac{\sum (\hat{y} - \bar{y})^2}{\sum (y - \bar{y})^2}$$

$$R^2_{\text{adjusted}} = \frac{1 - (1 - R^2)(N - 1)}{N - P - 1}$$

P = no. of features / predictors

N = total no. of samples / data points

X — X — X —

— X — X — X — X — X — X — X —

Bedrooms → Price

$P = 1 \rightarrow R^2 = 85\%$

Bedrooms → Location → Price

$P = 2 \rightarrow R^2 = 90\% \uparrow \uparrow$

Gender → Bedrooms → Location → Price

$P = 3 \rightarrow R^2 = 91\%$

not correlated but still there might be increase in R^2

Krish Naik

$$R^2 = 1 - \frac{(1 - R^2)(N - 1)}{N - P - 1}$$

$P = 2 \quad R^2 = 90\% \quad R^2_{\text{adjusted}} = 86\%$

$P = 3 \quad R^2 = 91\% \quad R^2_{\text{adjusted}} = 82\%$

when P is increasing $N - P - 1$ is decreasing and $(1 - R^2)(N - 1)$ will be bigger number. So $1 - \text{bigger number}$ is always decreasing

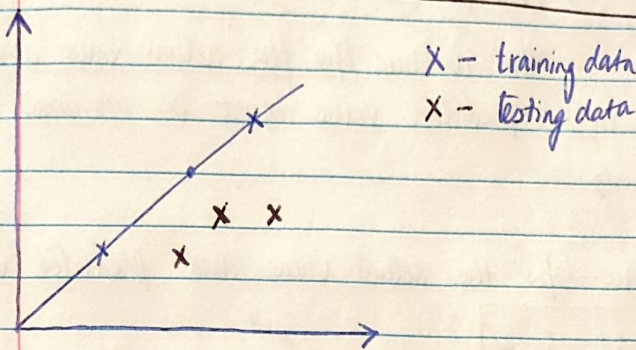
$$P = 2 \gg N - P - 1 \gg P = 3$$

as P is increasing there will be R^2_{adjusted} lesser

Squared Error Function

$$\text{Cost function} = \frac{1}{2m} \sum_{i=1}^m (h_0(x^i) - y^i)^2$$

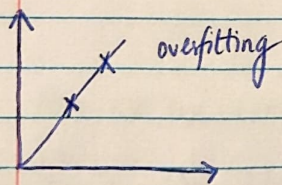
$$J(\theta_0, \theta_1)$$



Overfitting $\Rightarrow$ model performs well $\rightarrow$ training data (Low Bias)
fails to perform well $\rightarrow$ testing data (High variance)

Underfitting $\Rightarrow$ model accuracy is bad $\rightarrow$ training data (High Bias)
model accuracy is bad $\rightarrow$ testing data (High variance)

Model 1	Model 2	Model 3
Training accuracy = 90%	Training acc = 92%	Training acc = 70%
testing accuracy = 80%	testing acc = 91%	testing acc = 65%
↓	↓	↓
overfitting	Generalised model (Low bias, Low variance)	Underfitting



$$\begin{aligned} J(\theta_1) &= 0 \\ &= \frac{1}{2m} \sum_{i=1}^m (h_0(x^i) - y^{(i)})^2 \\ &= (\hat{y}^i - y^i)^2 \\ &= 0 \end{aligned}$$

So, Ridge & Lasso come into picture when there is overfitting

Ridge Regression (L2 Norm)

$$\rightarrow \text{cost function} = (h_0(x^{(i)}) - y^{(i)})^2 + \lambda (\text{slope})^2$$

$\rightarrow$ purpose: prevents overfitting

$\lambda \rightarrow$ hyperparameter

Lasso Regression (L1 Regularization)

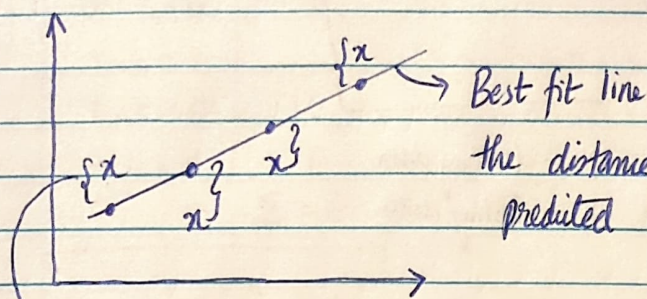
$$\rightarrow \text{cost function} = (h_0(x^{(i)}) - y^{(i)})^2 + \lambda |\text{slope}|$$

$\rightarrow$ purpose: prevent overfitting and has feature selection

$$h_0(x) = \hat{y} = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_3 \dots + \theta_n x_n$$

$$\rightarrow |\theta_0 + \theta_1 + \theta_2 + \dots + \theta_n|$$

Hypothesis: $h_0(x) = \theta_0 + \theta_1 x$



the distance b/w the actual value and predicted value must be minimal

eqⁿ for the distance b/w the actual value and predicted value

Cost function $\Rightarrow$

$$J(\theta_0, \theta_1) = \frac{1}{2m} \sum_{i=1}^m (h_0(x^i) - y^i)^2$$

predicted value
actual value

we are dividing this by $\frac{1}{2m}$ for the derivation purpose

in order to get the best fit line we need to change the θ_0 & θ_1

$$\frac{d(x^2)}{dx} = 2x$$

So we have $\frac{1}{2}$ to cancel 2

$$\frac{1}{2} \times 2x = x$$

what we need to solve?

minimize $\frac{1}{2m} \sum_{i=1}^m (h_0(x^i) - y^i)^2$
 θ_0, θ_1

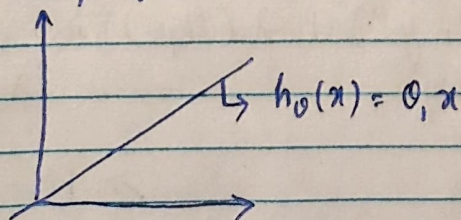
$\Downarrow$

$J(\theta_0, \theta_1) \rightarrow$ we need to minimize this

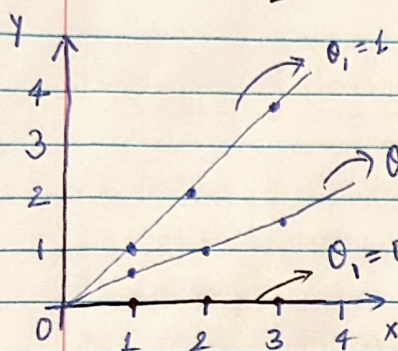
* $h_0(x) = \theta_0 + \theta_1 x$

if $\theta_0 = 0 \rightarrow$ then the line passes through origin where y coordinate is 0

$\uparrow$ y-intercept
 $\rightarrow$ slope / gradient
 $\Rightarrow y = c + mx$



$$m = \theta_1 = \frac{y_2 - y_1}{x_2 - x_1}$$



$$h_0(x) = \theta_1 x + \theta_0 \quad (\because \theta_0 = 0)$$

$$= \theta_1 x + 0$$

$$= \theta_1 x$$

actual values
 $\begin{Bmatrix} (1,1) \\ (2,2) \\ (3,3) \end{Bmatrix}$

$$J(\theta_1) = \frac{1}{2m} \sum_{i=1}^3 (h_0(x^i) - y^i)^2$$

$$= \frac{1}{2m} [(1-1)^2 + (2-2)^2 + (3-3)^2]$$

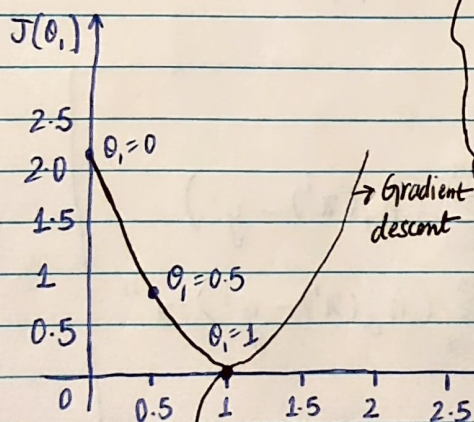
$$= \frac{1}{2m} [0]$$

Case 1 $\rightarrow$ where $\theta_1 = 1$

$$J(\theta_1) = 0$$

when $\theta_1 = 1$ then $J(\theta_1) = 0$

Cost function



This point is called as Global Minima

Case 2 $\rightarrow$ when $\theta_1 = 0.5$

$$J(\theta_1) = \frac{1}{2m} \sum_{i=1}^3 (h_0(x^i) - y^i)^2$$

$$= \frac{1}{2m} [(0.5-1)^2 + (1-2)^2 + (1.5-3)^2]$$

$$= \frac{1}{2 \times 3} [0.25 + 1 + 2.25]$$

$$= \approx 0.58$$

Case 3 $\rightarrow$ when $\theta_1 = 0$

$$J(\theta_1) = \frac{1}{2m} \sum_{i=1}^3 (h_0(x^i) - y^i)^2$$

$$= \frac{1}{6} [(0-1)^2 + (0-2)^2 + (0-3)^2]$$

$$= \frac{1}{6} \times [1 + 4 + 9]$$

$$\approx 2.3$$

Here we use Converge Algorithm for updating θ_1 and to achieve global minima

Repeat until Converge

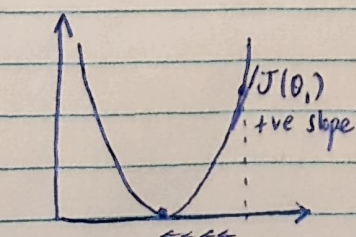
$$\theta_j := \theta_j - \alpha \left[\frac{\partial}{\partial \theta_j} J(\theta_0, \theta_1) \right]$$

derivative/slope

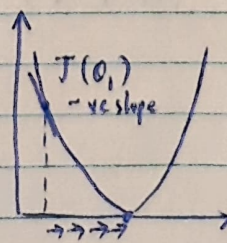
$\alpha \rightarrow$ Learning Rate

usually $\alpha = 0.01$

There is no local minima in LR with respect to gradient descent



$$\theta_1 := \theta_1 - \alpha(+ve)$$



$$\theta_1 := \theta_1 - \alpha(-ve)$$

$$\theta_1 := \theta_1 + \alpha(+ve)$$

Gradient descent Algorithm.

repeat until convergence

{

$$\theta_j := \theta_j - \alpha \frac{\partial}{\partial \theta_j} J(\theta_0, \theta_1)$$

}

J=0 and 1

$$\frac{\partial}{\partial \theta_j} J(\theta_0, \theta_1) = \frac{\partial}{\partial \theta_j} \frac{1}{2m} \sum_{i=1}^m (h_\theta(x^i) - y^i)^2$$

$$h_\theta(x) = \theta_0 + \theta_1 x$$

$$\checkmark \frac{\partial}{\partial \theta_0} = 0$$

$$j=0 \Rightarrow \frac{\partial}{\partial \theta_0} J(\theta_0, \theta_1) = \frac{1}{m} \sum_{i=1}^m (h_\theta(x^i) - y^i)$$

$$j=1 \Rightarrow \frac{\partial}{\partial \theta_1} J(\theta_0, \theta_1) = \frac{1}{m} \sum_{i=1}^m (h_\theta(x^i) - y^i) x^{(i)} \quad \checkmark \frac{\partial}{\partial \theta_1} = x$$

Repeat until convergence

{

$$\theta_0 := \theta_0 - \alpha \frac{1}{m} \sum_{i=1}^m (h_\theta(x^i) - y^i)$$

$$\theta_1 := \theta_1 - \alpha \frac{1}{m} \sum_{i=1}^m (h_\theta(x^i) - y^i) x^{(i)}$$

}

Performance Metrics

R^2 and Adjusted R^2

$$R^2 = 1 - \frac{SS_{Res}}{SS_{Total}} = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}$$

$$\hat{y}_i = h_\theta(x)$$

$$R^2_{adjusted} = 1 - \frac{(1 - R^2)(N - 1)}{(N - p - 1)}$$

Assumptions of Linear Regression

- 1> Normal | Gaussian Distribution $\rightarrow$ Model will get trained well.
- 2> Standardization { Scaling data } $\rightarrow$ Z-score $\mu=0, \sigma=1$
- 3> Linearity
- 4> Multi Collinearity